Module 13: How Populations Evolve — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Define microevolution in terms of allele frequencies.
2. Describe the five mechanisms of microevolution.
3. Contrast the "Bottleneck Effect" and the "Founder Effect."
4. Visually identify the three types of natural selection (Directional, Stabilizing, Disruptive).
5. Grasp the conceptual role of the Hardy-Weinberg equilibrium.

The Big Picture

> Natural selection is not the only way a population can evolve! Evolution simply means a change in a population's genetic makeup over generations. While natural selection is the only mechanism that consistently leads to *adaptation*, other powerful forces—like pure random chance—can dramatically alter a population's evolutionary path.

Key Terms & Concepts

1. The Five Fingers of Microevolution

Microevolution is defined specifically as a change in allele frequencies in a population over successive generations. There are five main ways this happens:

1. **Mutation:** Random changes in DNA create brand new alleles. This is the ultimate source of all genetic variation.
2. **Gene Flow (Migration):** The movement of individuals (and their alleles) into or out of a population. (Example: pollen blowing to a new valley).

3. **Genetic Drift (Chance):** Random fluctuations in allele frequencies. This affects *small populations* the most (detailed below).
4. **Non-random Mating:** When individuals prefer mates with particular traits (sexual selection), driving certain alleles up in frequency.
5. **Natural Selection:** Environmental pressures favor traits that increase survival and reproduction. (The only one that results in targeted adaptation).

2. Genetic Drift: Bottleneck vs. Founder Effect

Genetic Drift is evolution driven by random chance instead of survival of the fittest. It has two main varieties:

- **The Bottleneck Effect:** A catastrophic event (like a wildfire or hurricane) indiscriminately kills off a massive portion of the population. The few survivors rebuild the population, but massive genetic diversity is permanently lost.
- **The Founder Effect:** A small group of individuals breaks off from a larger population to establish a new colony (like birds landing on a remote island). The new population's gene pool is limited only to the genes of those few "founders".

3. Three Modes of Natural Selection

When natural selection acts on a trait, it can shift the population's physical traits (phenotypes) in three main ways, easily visualized as bell curves:

- **Directional Selection:** Shifts the overall makeup of the population by favoring variants at ONE extreme of the distribution. (Example: Darker mice survive better among dark volcanic rocks).
- **Stabilizing Selection:** Removes extreme variants from the population and preserves the intermediate (middle) types. (Example: Human birth weights—too small is dangerous, too large is dangerous; average is safest).
- **Disruptive Selection:** Favors variants at BOTH extremes of the distribution, while intermediate types are selected against. (Example: Finches with large beaks crack tough seeds, small beaks get tiny seeds, but medium-beak finches struggle with both).

4. Hardy-Weinberg Equilibrium

If none of the above five evolutionary forces are acting on a population, the population will simply not evolve. Its allele frequencies will stay perfectly stable generation after generation. We call this non-evolving state the **Hardy-Weinberg equilibrium**. In reality, populations are pretty much always experiencing at least one evolutionary pressure, so they are rarely in perfect equilibrium. Scientists use Hardy-Weinberg as a mathematical baseline to measure exactly *how fast* a population is evolving.

Study Tips

1. **Don't confuse gene flow and genetic drift.** Gene flow relies on *flow* (migration/movement between groups). Genetic drift relies on *drifting* (random luck of the draw).
2. **Natural selection is adaptive; Drift is not.** Bottlenecks don't care if an animal is strong, fast, or smart—if the hurricane lands on them, they die. Drift is about luck, selection is about adaptation.
3. **Draw the modes of selection.** When studying Directional, Stabilizing, and Disruptive selection, draw three simple bell curves and shade in the areas that represent the individuals surviving better.